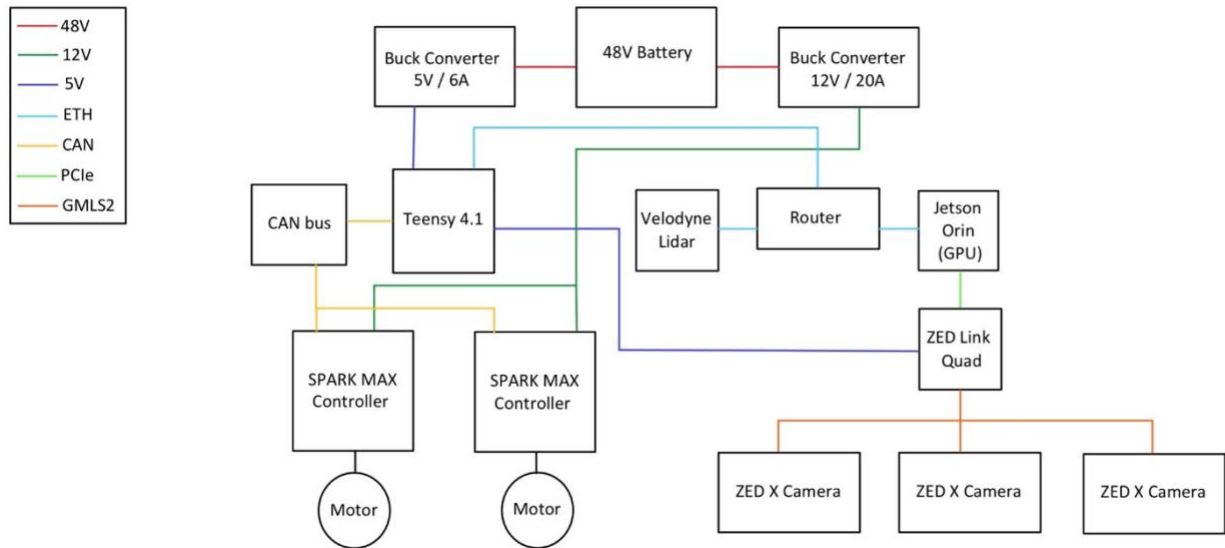


Electrical Design



4.1 Overview

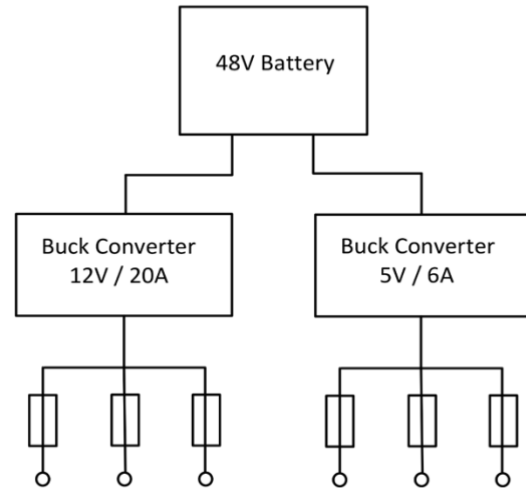
For our 2026 electrical design, we utilized a system of components for power distribution, network and computation, sensing, and motor control. The sensing system consists of three cameras and a LiDAR unit, and a GPS module, which interface with our Jetson GPU. The Jetson serves as the primary computation unit, processing sensor data and transmitting control data to a Teensy 4.1 microcontroller. The Teensy is responsible closed-loop motor control, communicating with SPARKMAX motor controllers over a shared CAN bus to ensure reliable and coordinated movement. Power is supplied by a 48V battery and stepped down to 12V and 5V rails using buck converters to safely power the rest of the system. The emergency stop system is implemented in hardware by directly interrupting power to the SPARKMAX controllers, ensuring immediate shutdown of all motor outputs when triggered.

4.2 Power System

The power distribution system uses a 48V battery connected to two buck converters that step the voltage down to required levels for different subsystems, through a power switch. The first converter regulates the voltage to 12V with a maximum output of 20A, providing sufficient current for the motor systems and other high-power components. The second

converter steps the voltage down to 5V with a maximum output of 6A, which is used to power our Teensy 4.1 microcontroller.

To ensure electrical safety and protect against overcurrent conditions, both power rails are distributed through fused connections. The 12V rail is divided across five individual fuses and the 5V rail includes a dedicated fuse for the Teensy. This structured approach to power distribution improves fault isolation and enhances overall system reliability.



4.3 Motor Control

The motor and feedback control system utilizes two SPARK MAX motor controllers to drive NEO brushless motors. Both controllers are connected over a shared CAN bus transceiver, enabling efficient communication with the Teensy 4.1 microcontroller. The SPARK MAX controllers provide real-time feedback through the integrated motor encoders, allowing the system to monitor speed and position. This feedback, combined with data from the sensing system, is used by the Teensy to implement closed-loop PID control for precise regulation of motor behavior. By integrating both encoder and sensor data, the system achieves improved accuracy and responsiveness in maintaining desired speeds and positions.

4.4 Sensing

This system incorporates three ZED X stereo cameras, which are connected to a ZED Link Quad using the GMSL2 communication protocol. This setup enables high-bandwidth, low-latency image transmission to the onboard Jetson GPU for real-time processing. In addition to visual sensing, a Velodyne LiDAR unit is used to capture precise 3D spatial data of the surrounding environment, enhancing obstacle detection and mapping capabilities. An Xsens GPS module is also integrated to provide accurate positioning and orientation data.

4.5 Network and Computation

Our network and computation system utilizes a router that facilitates communication between all major processing and sensing components. The Jetson GPU serves as the primary computation unit, handling vision processing and object detection. The ZED Link Quad interface is directly connected to the Jetson through PCIe, ensuring high-speed data transfer from the ZED X cameras. The router handles communication between the Velodyne LiDAR, Teensy 4.1, and Jetson GPU using Ethernet, while the Xsens GPS connects directly to the Jetson GPU. The router further transmits system data to a remote laptop over Wi-Fi, which allows for monitoring and remote operation. This network architecture enables efficient communication between sensing and computation ensuring coordinated system performance.